MILESTONE 2
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UW RENCANA

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1.0 / INTRODUCTION

UW Rencana is an application created with the intention of facilitating the house finding process in Stratford for upper year students in the University of Waterloo's Global Business and Digital Arts program. Designed by GBDA students at the University of Waterloo (*Zaid Amer, Gregory Cotton, Celina Nguyen and Helen Chen*), our team knows and has seen firsthand the issues students will go through to find housing in Stratford. This Problem will only be amplified by the future changes to the program, involving students moving permanently to the Stratford Campus at an earlier date (second year as opposed to third year).

The issues of a large number of students moving to Stratford, the lack of affordable student housing in Stratford, finding roommates, and even just the general limited knowledge most students have of Stratford collectively make house hunting a taxing process. As a team, UW Rencana has tasked ourselves with creating a phone application designed to connect students with prospective roommates, houses, and apartments in the Stratford area to help take a little stress out of University.

An application that is totally student run as such will foster a good community between students enrolled in Global Business and Digital Arts who are living, commuting to, and looking for housing in Stratford. This report will outline the high-fidelity prototype we have developed, as well as it will cover user testing as results we have collected.

2.0 / HIGH FIDELITY PROTOTYPE

Sections **2.0** though **2.2** discusses the final design of our application, the process that we undertook to achieve this design, and the details the UW Rencana high-fidelity prototype. We will create a help document to explain the various features that UW Rencana boasts as well as to help any new and potentially struggling users navigate their way around our easy-to-use and visually appealing interface.

2.1 / DESIGN DOCUMENT

As we developed our application, our outlook on the usability goals we should set for UW Rencana shifted. Though they did remain mostly the same, we adapted some slight changes for the better of the application.

A primary usability goal that did remain the same was the principal to keep this application organized, simple, but effective. This was a goal that was broken down and dealt with in subsections to better manage it. Keeping the application focused and organized for students was done with the intention of creating a more seamless and streamlined experience for our users. Housing was broken down into sub-categories, listing ranges in price that students were willing to pay (price per month) as well as the different housing types, as can be seen in *Img 1* (more detail on the filtering system can be found in **2.2 Help Documentation**). By allowing the students to filter out housing styles they find unappealing and price points they find unaffordable, we will create a much more pleasant and efficient experience. Removing this extra hassle enhances our application, the user experience, and the likelyhood of success in terms of finding affordable student living or roomates.

Img 1

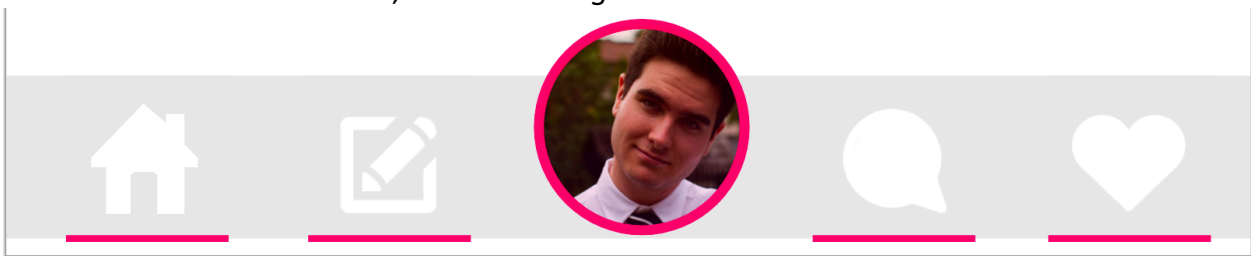
The method we used to achieve this goal of ours was through dropdown menus, which can be found at the starting of the application, or when a user makes a new account. By adding this feature directly at the starting of UW Rencana, we are able to determine the user's preferences as soon as we can to ensure that our application wastes no time in narrowing down viable housing options.

A second goal we set for ourselves was the option for students and users to review landlords. This was one of the most important goals we set when developing this idea, as we strongly believe no student should have to struggle with a hard-to-handle landlord. Leaving reviews gives the students the opportunity to not only avoid landlords who have poor track records, but also the opportunity to help warn future tenants about certain individuals. UW Rencana strives not only to help students house-hunt but also to foster a friendly and tight knit community within GBDA, and our team believes this feature is a huge step in helping build that sense of community. Our application features a "new post" button in which

users can select what type of post they'd like to write up; among those options will be the option to review a landlord. We have implemented a star classification system which gives users the option to rate landlords on a scale of one to five stars, and the option to leave comments meant to justify the rating given.

As our overall vision is to facilitate the house-finding process, a goal that UW Rencana holds at high importance is the option to message other users directly and privately through the application. This eliminates the extra, and unnecessary, step of users using a third-party application or website (such as email or various social media/messaging platforms) to communicate with others about living arrangements. We have designed a messaging interface which is easy to use and intuitive; by mimicking the simple interface shared by popular and already established messaging platforms, UW Rencana can ensure that a vast majority of users will already be familiarized with the messaging layout, and that those who aren't will have to expend very little effort in learning their way around. An intuitive messaging system in our applications will be useful when users try to coordinate anything; UW Rencana will be an all-in-one application for Stratford house-hunting, all while maintaining a high level of organization and an easily navigable interface.

Making public posts advertising housing, or the lack thereof, as well as requesting roommates was another feature UW Rencana highly values, as it is the primary way for students to find housing or roommates and it adds to the sense of community we'd like to foster. The option to write a new post will be available on the menu bar, found at the bottom of the user's screen, as seen in *Img 2*.



Img 2

A usability goal our team added in after low-fidelity prototyping was to enhance the security of the application and restrict the number of joke posts we could potentially receive from students. We carried this goal out by requiring login to be done through your Waterloo Quest account, using student number and email. This feature will not only allow us to keep students not in the university out of the application; logging in with Waterloo credentials including student number will allow UW Rencana to user information, such as name, program, and any other relevant details, which only lessens the information our users have to enter manually, once again increasing our application's efficiency.

The information used on the high-fidelity prototype for the sample profile of Samuel Foss on UW Rencana is in part improvised, however many aspects were drawn

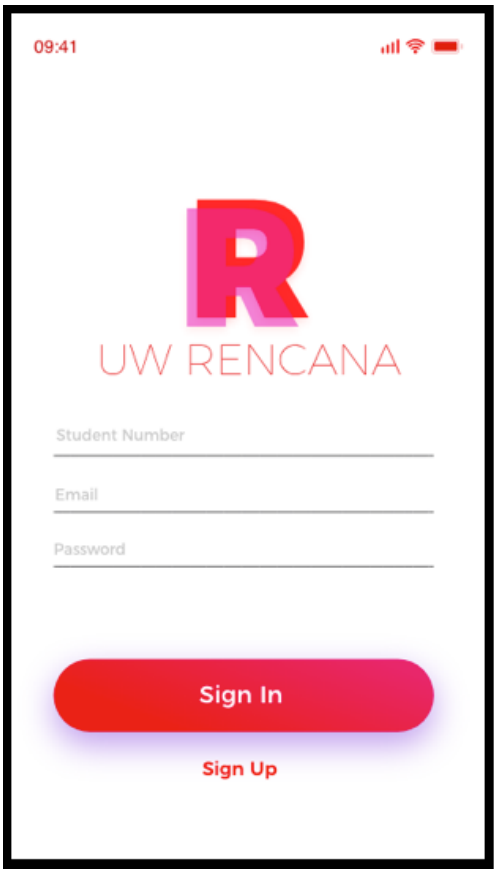
from Samuel Foss, a member of our usability tests. Some information has been altered for privacy and all used information was done so with consent of Samuel Foss.

2.2 / HELP DOCUMENTATION

To assist users who may struggle we have created the following manual giving a rundown on the UW Rencana application. This document can also be used as an outline for the application's high fidelity-prototype.

When opening the application the user is greeted with this screen to welcome the user to the app.

Figure 1



Upon transgressing from the previous screen the user is presented with a login screen. With this they have two options, input their data and sign in, or if they have not used the app before sign up.

Figure 3.

After creating an account the user is given this welcome message. The message is followed up by multiple screens, similar to this (listed below), filled with a sort of questionnaire, in which one inputs their personal housing preferences. The preferences consist of; style of residence, number of roommates, budget and a new feature called tags. Tags is a feature in which one can use a hashtag to denote a particular feature they are looking for within a particular residence in order to further enhance the home searching experience and find the perfect house for oneself

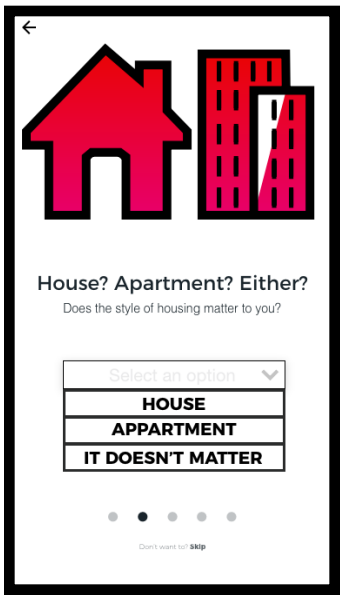


Figure 4.

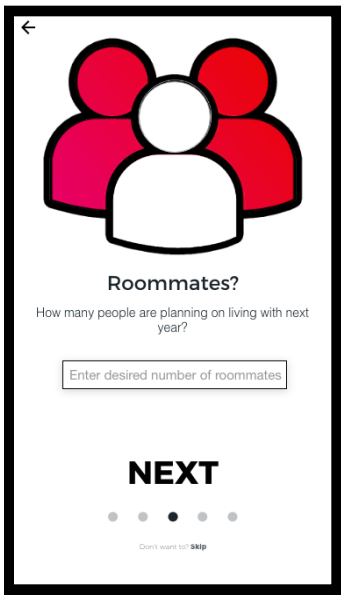


Figure 5.

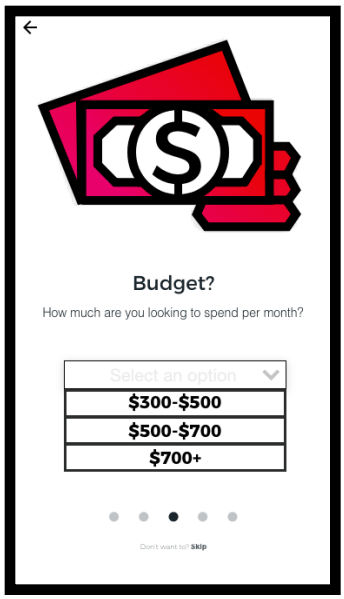


Figure 6.

Figure 6.

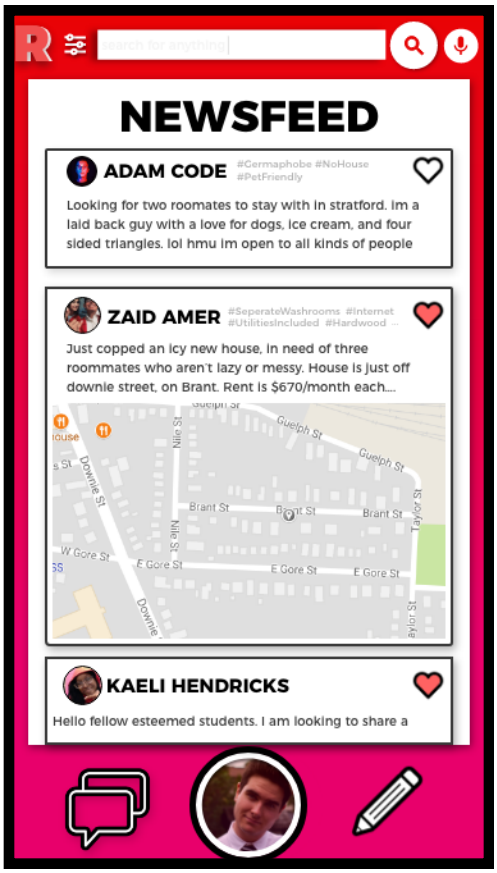


Figure 7

Figure 8

Once both the initial “quiz” and account creation is done one is then presented with the home screen (Figure 7). When on the home screen the the user is given a plethora of options. The top bar presents a search bar which is used to search for specific posts in your newsfeed. The search bar is usable by touch, or the microphone to the right of it. To the left of the search bar, is a slider, which can be used to access settings. On the bottom there are three major icons, the chat bubbles open the messages, the profile picture opens up ones personal profile and the pencil opens up a post page. Once on the posting screen (figure 8) the user is given a box to type within, as well as the option to add a picture, add a type of post (roommate searching, housing post etc.) as well as adding tags. To continue the user can either press the post button (the paper airplane) or click outside of the box to return to the previous screen.

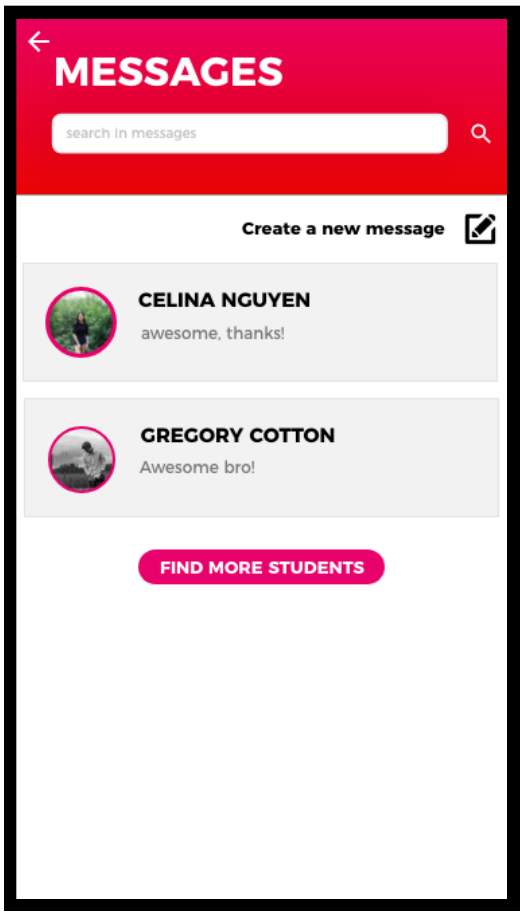


Figure 9

When clicking the chat bubble aforementioned chat bubble icon, the user moves to this main messages page (figure 9). On the page one is given a couple options. The first is to search within your messages to either find a specific message, or a person. One could find more friends button (figure 11), or click pencil icon to message someone new (figure 12).

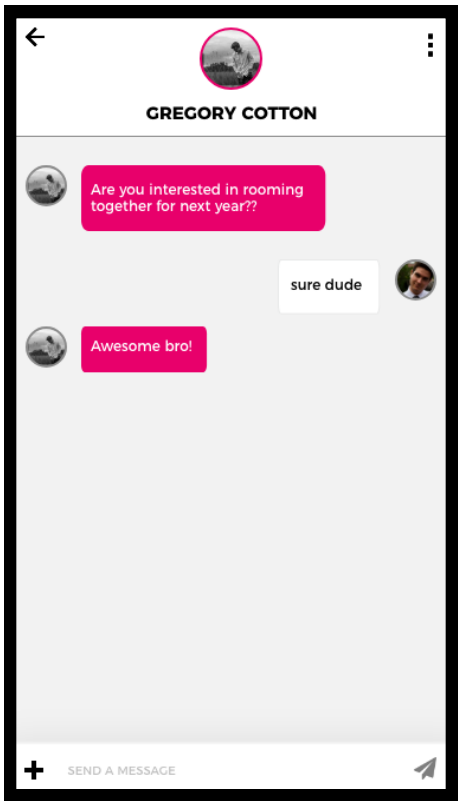


Figure 10

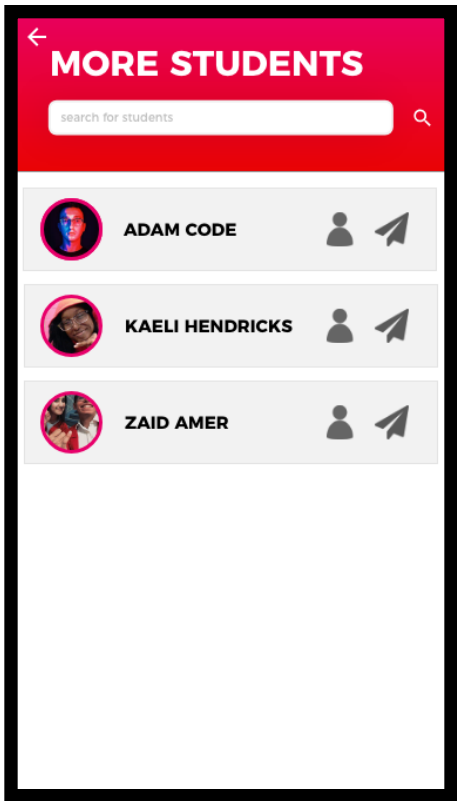


Figure 11

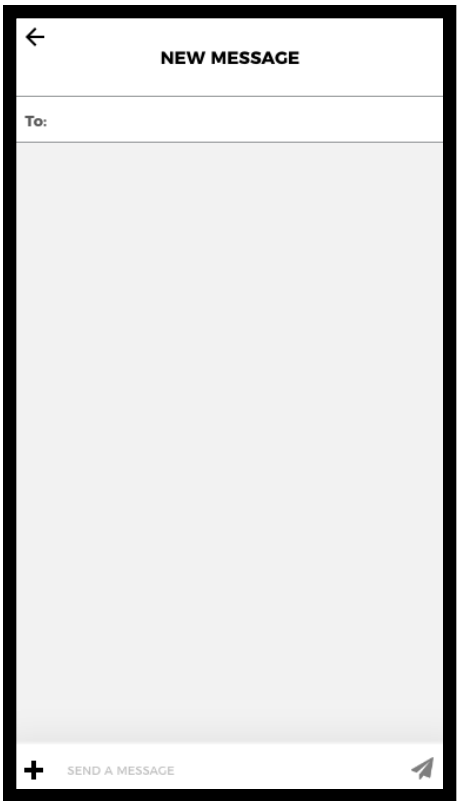


Figure 12

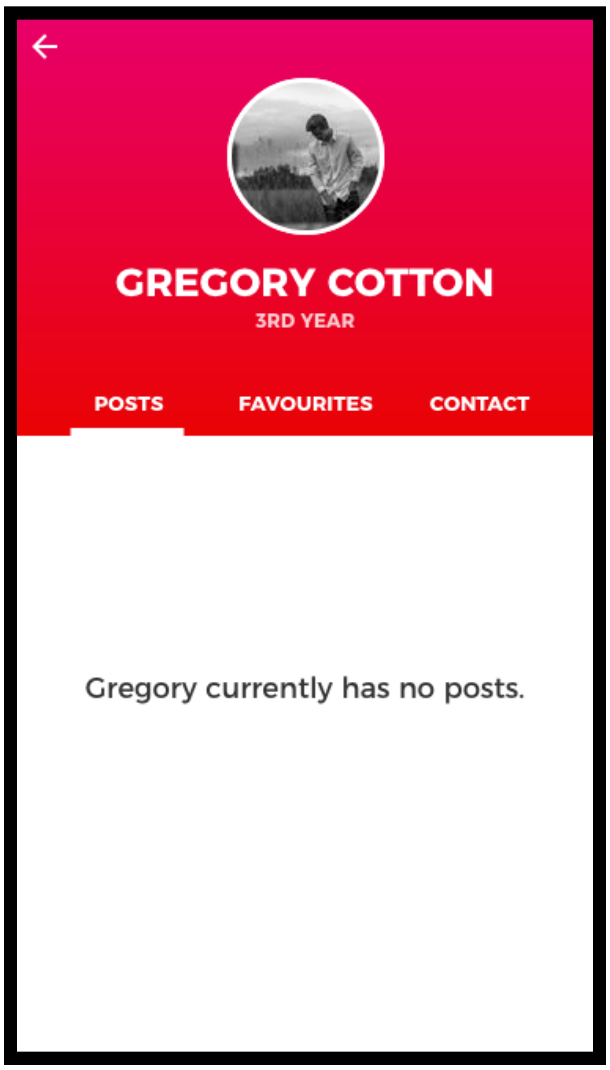


Figure 13

When clicking on the profile pictures within the messages section, the user is presented with whomever was clicked's profile (figure 13). The profile has three sections, each accessible by a swipe or press; posts, favourites (figure 14) and contact (figure 15). The posts page shows the persons posts, the favourites their favourited post and the contact always shows the person's email, and phone number if chosen to share.

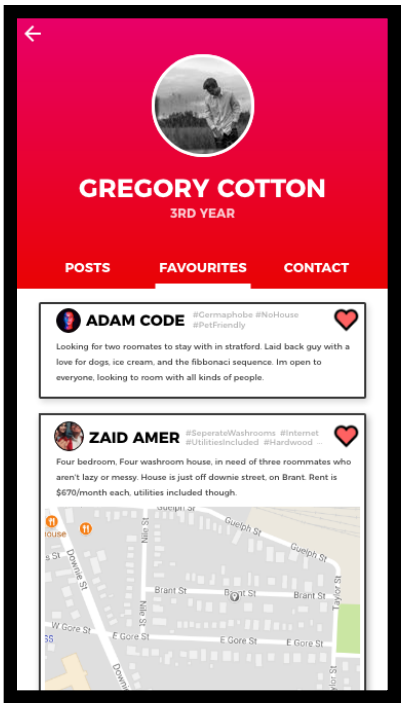


Figure 14

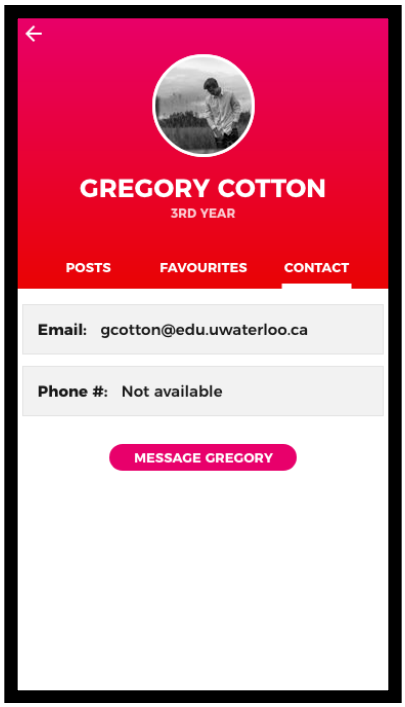


Figure 15

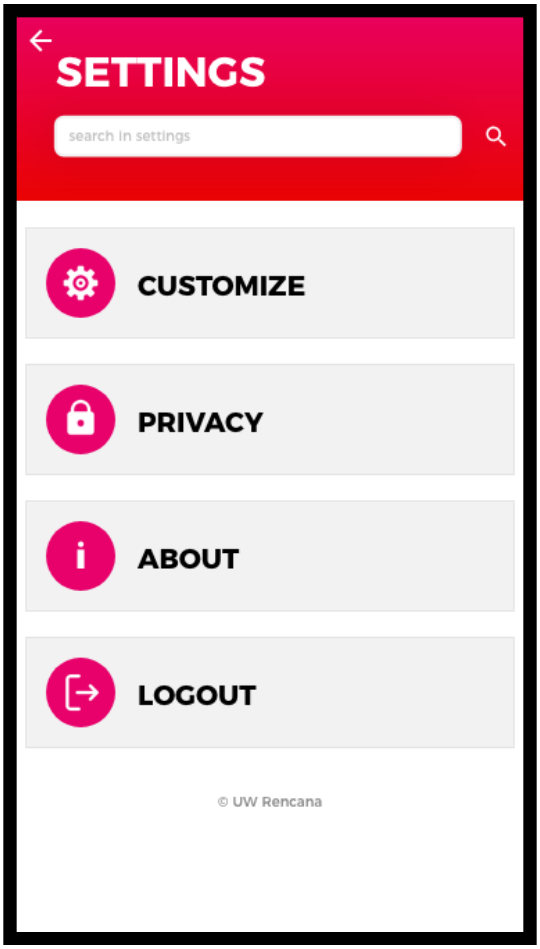


Figure 16

When the uses access's the settings page (off either their personal profile or the home screen) they are presented with a new screen (figure 16). On that screen one has several options. One can check the about section, containing a small paragraph listing the purpose of Rencana, or logout from here. Additionally the user can edit privacy (figure 17) as well as being able to customize certain options (figure 18) such as profile picture, as well as housing settings (figure 19)

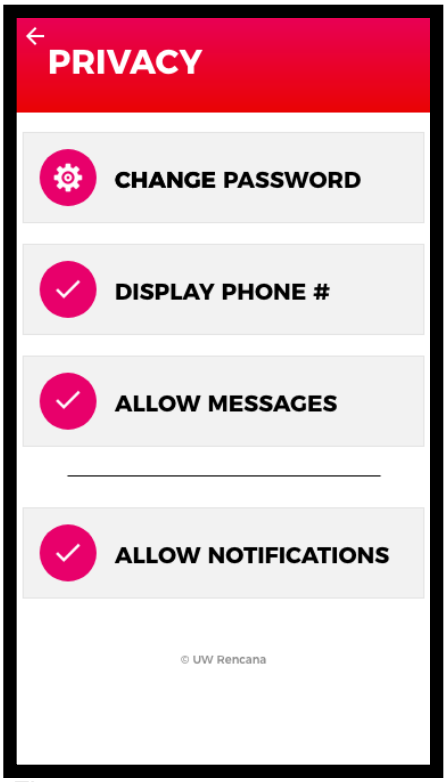


Figure 17

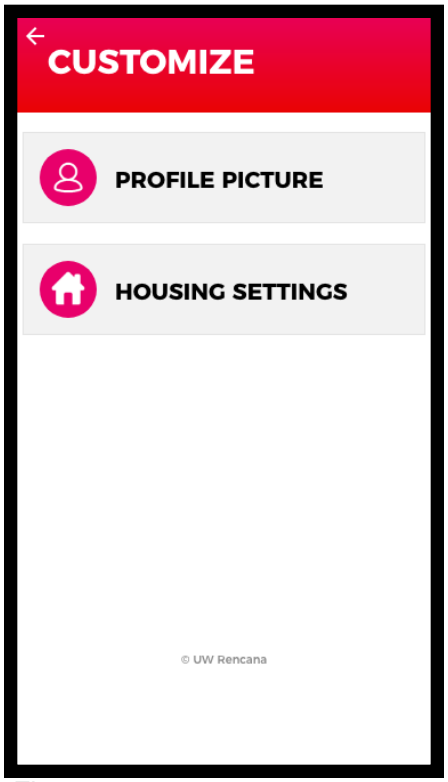


Figure 18

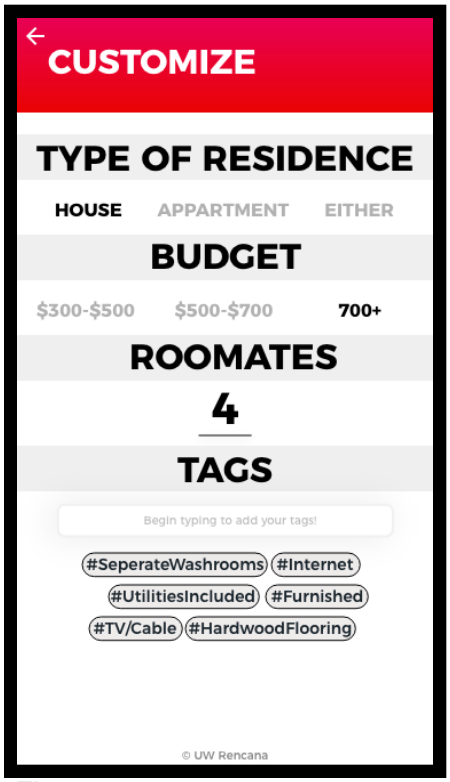


Figure 19

3.0 / USER STUDY PLAN

Section **3.0** will focus UW Rencana's usability testing: specifically, on the questions users will have to answer as they progress through our usability testing. We will then go on to comparing and analyzing our collected data from testing. Our team also plans on conducting a few questionnaires to fully grasp how users felt about and reacted to UW Rencana.

Section **3.1** will display the tasks we will have users complete during usability testing. From here, we will analyze the gathered information in Section **3.2**, and then show questionnaire questions and results in Section **3.3**.

3.1 / METHODOLOGY

Step #1 – Setting Up

- Please create an account;
- Complete the series of dropdown menus you should see;
- Visit your profile through the menu bar at the bottom of your screen;
- Using the settings option on your profile, go back and revise all the information you entered throughout the series of dropdown menus.

Step #1 was designed with the intention to encourage users to get started on UW Rencana and set up filters specific to their housing preferences.

Step #2 – Favorited Posts

- Please select a user from your newsfeed, and view their profile;
- Select the favorites tab on a user's profile to view their favorites;
- Select a post that you selected user has added to their favorites and view that post individually;
- Using the menu bar at the bottom of your screen, return to your newsfeed;
- Select the favorites tab on your profile to view your favorited posts;

Step #2 will test the user's ability to interact with posts and profiles found on their newsfeed, as well as favorited posts by users and themselves. This is a fairly simple yet very important process.

Step #3 – Messaging Users

- From the bottom menu bar, select the option to compose a new message;
- Select a user to send a message to from a list of recommended users;

While short, Step #3 tests out the user's ability to find and utilize the messaging feature. Due to time constraints, we are unable to create a prototype that will actually sends off messages to users, hence the reason this step stops after selecting a recipient.

Step #4 – Write a Post/Review

- Please select the option to create a new post;

- Select the option to review a landlord;

Writing posts and reviews in potentially one of the most important features in UW Rencana, as it is what allows students to make initial communication by sharing their need for housing/roommates/landlord reviews before they use the messaging feature. Step #4 is relatively brief, but will task the users with trying to create a new review from scratch.

Step #5 – Hashtags

- Return to your home newsfeed;
- Select a post in your feed;
- Click one of the hashtags to view similar posts;

Step #5 is quick, but will have users test out the hashtag feature, which will be used to narrow search results down even more.

3.2 / MEASURES AND METRICS

Section **3.2** will cover the measures and metrics that our team will be using while conducting usability testing and in both pre-test and post-test questionnaires.

3.2.1 / MEASURES

Section **3.2.1** will discuss how our team plans to measure user reactions to our application's high-fidelity prototype during usability testing. All data gathered is primary and was collected directly from our post-test questionnaire we gave all users who tested for us. By fully understanding how our subjects responded to UW Rencana, we are better able to add or phase out certain features entirely.

3.2.2 / METRICS

Our team's metrics will be collected through the likert-scale questions that can be found in our post-test questionnaire, as well as through the short answers. We were looking to measure not only how visually appealing users found our application, but how much they think our application needs to be further developed and changed before it is a fully and perfectly useable application.

3.3 / QUESTIONNAIRES

We gave each individual tested a set of questionnaires to complete; one before they used the UW Rencana application, and one to complete afterwards. The following section will show the questions asked, both prior to and after using the application, and the results.

3.3.1 / PRE-TEST QUESTIONNAIRE

The questionnaire below was given to usability test subjects before they interacted with the UW Rencana high-fidelity prototype.

PRE-TEST QUESTIONNAIRE: UW RENCANA

Q1 – What year are you in at the University of Waterloo?

a) First year
b) Second year
c) Third year
d) Fourth year
e) Fifth year + / MDEI

Q2 – What is your gender?

a) Male
b) Female

Q3 – Do you/Do you wish to live in Stratford while studying there full time?

a) Yes, I am interested in living in Stratford
b) No, I would prefer to commute/other

Q4 – Do you have potential roommates to live with in upper years?

a) Yes
b) No
c) I don't plan on having roommates/I live at home

Q5 – In your opinion, what is the scariest part about looking for housing?
(short answer, multiple choice does not apply)

Q6 – Have you ever looked for housing before?

a) Yes
b) No

3.3.2 / POST-TEST QUESTIONNAIRE

The questionnaire below was given to usability test subjects after they interacted with the UW Rencana high-fidelity prototype.

POST-TEST QUESTIONNAIRE: UW RENCANA

Q1 – The User Interface for UW Rencana was…

a) Excellent
b) Good
c) Acceptable
d) Poor
e) Awful

Q2 – I found the visuals/graphics of UW Rencana…

a) Visually appealing; helpful
b) Decent; acceptable
c) Poorly made; confusing

Q3 - I would use UW Rencana to assist me when I look for houses;
Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree

Q4 – If you could add one feature to UW Rencana, what would you like to see?
(short answer, multiple choice does not apply)

Q5 – What was your biggest issue with the UW Rencana prototype?
(short answer, multiple choice does not apply)

a. How do you think we could resolve the issue?

Increase the text and perhaps make more space considering looking at large amounts of text on a small screen can be overwhelming

Q6 – In the beginning of UW Rencana when you sign up, do you find the initial filter-by options helpful?

a) Yes
b) No

Q7 – How helpful/important is student feedback to you?

a) Not important
b) Important

Q8 – Was there anything in particular that you liked about UW Rencana?

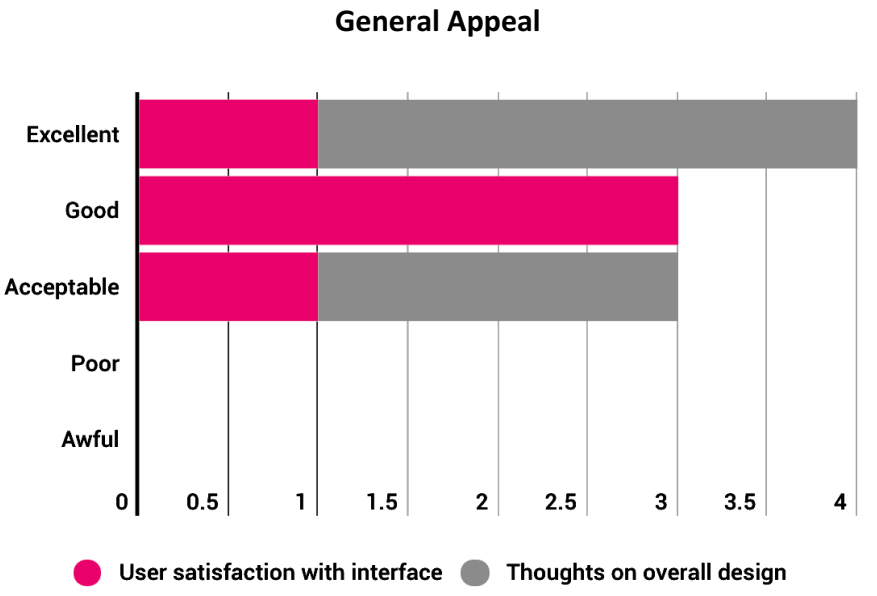
I really like being able to customize what I see on my newsfeed. The filter-by options were a great addition to the app.

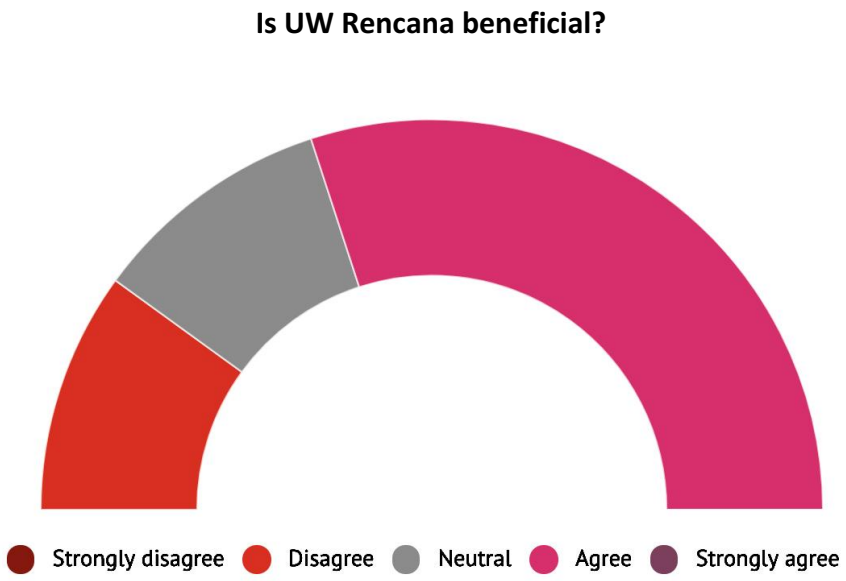
Q9 – Was there anything in particular that you didn't like about UW Rencana?

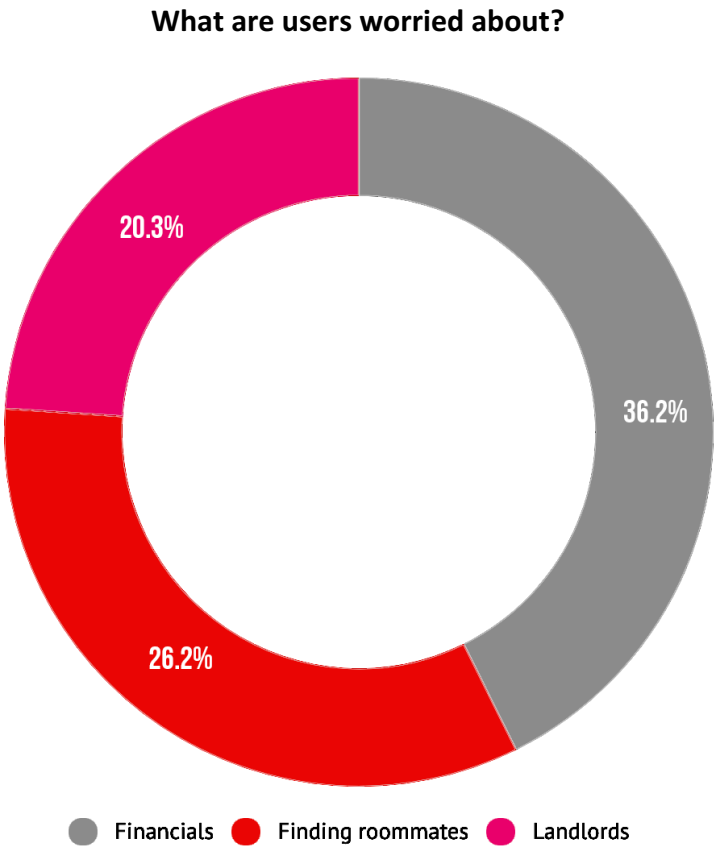
Nothing currently comes to mind

3.3.3 / RESULTS

This section will display results of the previous two questionnaires, which usability test subjects were asked to answer. Data is merely organized and shown here; for further analysis of quantitative and qualitative data see individual team members' **Milestone 2** papers.


General Appeal
Excellent
Good
Acceptable
Poor
Awful
0
0.5
1
1.5
2
2.5
3
3.5
4
User satisfaction with interface
Thoughts on overall design


Is UW Rencana beneficial?
Strongly disagree
Disagree
Neutral
Agree
Strongly agree


What are users worried about?
20.3%
36.2%
26.2%
Financials
Finding roommates
Landlords